THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{3}{4}$	Substitute the standard values
2	**(c)** 3	$\frac{c}{q}+\frac{p}{q}$
3	**(c)** $\frac{1}{2}$	$\frac{c}{p}-\frac{q}{p}$
4	**(b)** 2	Each ratio is 1
5	**(b)** $\frac{7}{3}$	Divide by $\cos\theta$
6	**(b)** 70°	$\cos(90^\circ-\theta)=\sin\theta$
7	**(a)** $\frac{7}{6}$	Substitute the standard values
8	**(c)** 5	$\frac{c}{q}+\frac{p}{q}$
9	**(a)** $\frac{2}{3}$	$\frac{c}{p}-\frac{q}{p}$
10	**(d)** 2	Each ratio is 1

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Divide by $\cos A$: $\frac{\tan A+1}{\tan A-1}$ (1) $=7$ (1).	1+1
12	$\frac{1}{2}=\sin 30^\circ$ (1); $2x+0=30$ (1); $x=15^\circ$ (1).	1+1+1
13	Use $\sin^2 45^\circ=\frac{1}{2}$, $\cos^2 45^\circ=\frac{1}{2}$ (2); value $=1$ (1).	2+1
14	Divide by $\cos A$: $\frac{\tan A+1}{\tan A-1}$ (1) $=\frac{17}{7}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{1}{2}=\sin 30^\circ$ (1); $3x+0=30$ (1); $x=10^\circ$ (1).	1+1+1
16	Hypotenuse $=\sqrt{15^2+8^2}=17$ (1); $\sin\theta=\frac{15}{17}$, $\cos\theta=\frac{8}{17}$ (1); $\sec\theta=\frac{17}{8}$ (1).	1+1+1
17	$\sqrt{\cos^2 A}=\cos A$ (1); $\cos A\cdot\sec A$ (1) $=1$ (1).	1+1+1
	OR: $\sqrt{3}=\tan 60^\circ$ (1); $2x+0=60$ (1); $x=30^\circ$ (1).	1+1+1
18	Hypotenuse $=\sqrt{24^2+7^2}=25$ (1); $\sin\theta=\frac{24}{25}$, $\cos\theta=\frac{7}{25}$ (1); $\sec\theta=\frac{25}{7}$ (1).	1+1+1
	OR: $\frac{\sin A}{\csc A}=\sin^2 A$, $\frac{\cos A}{\sec A}=\cos^2 A$ (2); sum $=1$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	LHS $=\left(\frac{1-\sin A}{\cos A}\right)^2$ (2); $=\frac{(1-\sin A)^2}{1-\sin^2 A}$ (1); $=\frac{1-\sin A}{1+\sin A}$ (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{3\tan A+2}{4\tan A+1}$ (1); $=\frac{3(\frac{15}{8})+2}{4(\frac{15}{8})+1}=\frac{61}{68}$ (2); verified with $\sin A=\frac{15}{17}$, $\cos A=\frac{8}{17}$ (1).	1+1+2+1
20	LHS $=\frac{\cos^2 A+(1+\sin A)^2}{\cos A(1+\sin A)}$ (2); $=\frac{2+2\sin A}{\cos A(1+\sin A)}$ (2); $=\frac{2}{\cos A}=2\sec A$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{2\tan A+5}{3\tan A+2}$ (1); $=\frac{2(\frac{24}{7})+5}{3(\frac{24}{7})+2}=\frac{83}{86}$ (2); verified with $\sin A=\frac{24}{25}$, $\cos A=\frac{7}{25}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin\times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta=\frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.

- Writing $\sin 2A = 2\sin A$ or $\sin(A+B) = \sin A + \sin B$, which are false.
- Wrong table values, e.g. $\sin 30^\circ = \dfrac{\sqrt{3}}{2}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{17}{5}$	Divide by $\cos\theta$
2	**(d)** 55°	$\cos(90^\circ-\theta)=\sin\theta$
3	**(b)** $\frac{3}{2}$	Substitute the standard values
4	**(b)** 4	$\frac{c}{q}+\frac{p}{q}$
5	**(b)** $\frac{3}{5}$	$\frac{c}{p}-\frac{q}{p}$
6	**(d)** 2	Each ratio is 1
7	**(c)** $\frac{23}{8}$	Divide by $\cos\theta$
8	**(b)** 42°	$\cos(90^\circ-\theta)=\sin\theta$
9	**(a)** $\frac{5}{4}$	Substitute the standard values
10	**(b)** 7	$\frac{c}{q}+\frac{p}{q}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\sqrt{3}=\tan 60^\circ$ (1); $3x+0=60$ (1); $x=20^\circ$ (1).	1+1+1
12	Use $\tan^2 60^\circ=3$, $\sin^2 30^\circ=\frac{1}{4}$ (2); value $=\frac{23}{4}$ (1).	2+1
13	Divide by $\cos A$: $\frac{\tan A+1}{\tan A-1}$ (1) $=\frac{31}{17}$ (1).	1+1
14	$\frac{1}{2}=\cos 60^\circ$ (1); $2x+10=60$ (1); $x=25^\circ$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{\sqrt{3}}{2}=\cos 30^\circ$ (1); $3x+0=30$ (1); $x=10^\circ$ (1).	1+1+1
16	Hypotenuse $=\sqrt{21^2+20^2}=29$ (1); $\sin\theta=\frac{21}{29}$, $\cos\theta=\frac{20}{29}$ (1); $\sec\theta=\frac{29}{20}$ (1).	1+1+1
17	$1+\tan^2 A=\sec^2 A$ (1); $\cos^2 A\sec^2 A$ (1) $=1$ (1).	1+1+1
	OR: $\frac{1}{\sqrt{3}}=\tan 30^\circ$ (1); $2x+0=30$ (1); $x=15^\circ$ (1).	1+1+1
18	Hypotenuse $=\sqrt{40^2+9^2}=41$ (1); $\sin\theta=\frac{40}{41}$, $\cos\theta=\frac{9}{41}$ (1); $\sec\theta=\frac{41}{9}$ (1).	1+1+1
	OR: $\sin^2 A\cdot\frac{\cos^2 A}{\sin^2 A}=\cos^2 A$ (1); $\cos^2 A\tan^2 A=\sin^2 A$ (1); sum $=1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	LHS $=\frac{\sec^2 A}{\csc^2 A}=\tan^2 A$ (2); RHS: $1-\cot A=\frac{\tan A-1}{\tan A}$ so $\frac{1-\tan A}{1-\cot A}=-\tan A$ (2); square $=\tan^2 A$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{5\tan A+3}{1\tan A+1}$ (1); $=\frac{5(\frac{21}{20})+3}{1(\frac{21}{20})+1}=\frac{165}{41}$ (2); verified with $\sin A=\frac{21}{29}$, $\cos A=\frac{20}{29}$ (1).	1+1+2+1
20	LHS $=(\sin^2\theta-\cos^2\theta)(\sin^2\theta+\cos^2\theta)$ (2); $=\sin^2\theta-\cos^2\theta$ (1); $=\sin^2\theta-(1-\sin^2\theta)=2\sin^2\theta-1$ (2).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{4\tan A+7}{2\tan A+1}$ (1); $=\frac{4(\frac{40}{9})+7}{2(\frac{40}{9})+1}=\frac{223}{89}$ (2); verified with $\sin A=\frac{40}{41}$, $\cos A=\frac{9}{41}$ (1).	1+1+2+1

Mistakes to avoid:

- Treating $\sin A$ as $\sin\times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta=\frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A=2\sin A$ or $\sin(A+B)=\sin A+\sin B$, which are false.

- Wrong table values, e.g. $\sin 30^\circ = \frac{\sqrt{3}}{2}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{3}{4}$	$\frac{c}{p} - \frac{q}{p}$
2	**(b)** 2	Each ratio is 1
3	**(d)** $\frac{31}{7}$	Divide by $\cos\theta$
4	**(d)** 78°	$\cos(90^\circ - \theta) = \sin\theta$
5	**(b)** 0	Substitute the standard values
6	**(c)** $\frac{5}{2}$	$\frac{c}{q} + \frac{p}{q}$
7	**(d)** $\frac{3}{7}$	$\frac{c}{p} - \frac{q}{p}$
8	**(b)** 2	Each ratio is 1
9	**(c)** $\frac{41}{20}$	Divide by $\cos\theta$
10	**(d)** 63°	$\cos(90^\circ - \theta) = \sin\theta$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use $\cos^2 30^\circ = \frac{3}{4}$, $\sin^2 60^\circ = \frac{3}{4}$ (2); value $= \frac{3}{2}$ (1).	2+1
12	Divide by $\cos A$: $\frac{\tan A + 1}{\tan A - 1}$ (1) $= -\frac{17}{7}$ (1).	1+1
13	$1 = \tan 45^\circ$ (1); $5x + 0 = 45$ (1); $x = 9^\circ$ (1).	1+1+1
14	Use $\sin^2 60^\circ = \frac{3}{4}$, $\cos^2 30^\circ = \frac{3}{4}$ (2); value $= \frac{3}{4}$ (1).	2+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{\sqrt{3}}{2} = \sin 60^\circ$ (1); $3x + 0 = 60$ (1); $x = 20^\circ$ (1).	1+1+1
16	Hypotenuse $= \sqrt{8^2 + 6^2} = 10$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1).	1+1+1
17	Use $(a-b)(a+b) = a^2 - b^2$ (1); $\csc^2 A - \cot^2 A$ (1) $= 1$ (1).	1+1+1
	OR: $\frac{1}{\sqrt{2}} = \cos 45^\circ$ (1); $5x + 0 = 45$ (1); $x = 9^\circ$ (1).	1+1+1
18	Hypotenuse $= \sqrt{12^2 + 9^2} = 15$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1).	1+1+1
	OR: $\frac{1}{\sec^2 A} + \frac{1}{\csc^2 A}$ (1); $= \cos^2 A + \sin^2 A$ (1) $= 1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	LHS $= \frac{1 + \frac{1}{\cos A}}{\frac{1}{\cos A}} = \cos A + 1$ (2); RHS $= \frac{1 - \cos^2 A}{1 - \cos A}$ (2); $= 1 + \cos A$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{6\tan A + 1}{3\tan A + 1}$ (1); $= \frac{6(\frac{4}{3}) + 1}{3(\frac{4}{3}) + 1} = \frac{9}{5}$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1).	1+1+2+1
20	LHS $= \sqrt{\frac{(1+\sin\theta)^2}{1 - \sin^2\theta}}$ (2); $= \frac{1 + \sin\theta}{\cos\theta}$ (2); $= \sec\theta + \tan\theta$ (1).	5
	OR: Divide numerator and denominator by $\cos A$ (1): $\frac{1\tan A + 8}{2\tan A + 1}$ (1); $= \frac{1(\frac{4}{3}) + 8}{2(\frac{4}{3}) + 1} = \frac{28}{11}$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1).	1+1+2+1

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- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta = \frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.

- Writing $\sin 2A = 2\sin A$ or $\sin(A+B) = \sin A + \sin B$, which are false.
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